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1 2.1 (l)

$U = \{1, 2\}$

$a = 2 \quad b = 1$

$P(1) = T \quad P(2)\ F$

$Q(1,1)$	$Q(1,2)$	$Q(2,1)$	$Q(2,2)$
F	T	F	T

$$(\exists x)((\forall y) \sim Q(y,x) \Rightarrow (P(x) \wedge (\exists z)(Q(z,x) \wedge Q(z,z))))$$

Case 1: let $x = 1$

1. $(\forall y) \sim Q(y,1)$

$= \sim Q(1,1) \wedge \sim Q(2,1)$

$= T \wedge T$

$= T$

2. $(\exists z)(Q(z,1) \wedge Q(z,2))$

$z = 1$; $Q(1,1) \wedge Q(1,1) = F$

$z = 2$; $Q(2,1) \wedge Q(2,2) = F$

Hence, $(\exists z)(Q(z,1) \wedge Q(z,2)) = F$

3. $P(1) \wedge (\exists z)(Q(z,1) \wedge Q(z,2))$

$T \wedge F = F$

4. $(\forall y) \sim Q(y,x) \Rightarrow (P(x) \wedge (\exists_z)(Q(z,x) \wedge Q(z,z)))$

$T \Rightarrow F = F$

Case 2: let $x = 2$

$$(\forall y) \sim Q(y,2)$$

Since $Q(1,2) = T$ $\quad Q(2,2) = T$

$\sim Q(1,2) = F$ $\quad \sim Q(2,2) = F$

Therefore $F \Rightarrow$ anything $= T$

Hence, the correct answer is True

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2.2 (c)

$\vdash (\forall x)\alpha(x) \vee (\forall x)\beta(x) \Rightarrow (\forall x)(\alpha(x) \vee \beta(x))$

$(\forall x)\alpha(x) \vee (\forall x)\beta(x)$

let a be arbitrary element of univ.

1.	$(\forall x)\alpha(x) \vee (\forall x)\beta(x)$	hypothesis
2.	$(\forall x)\alpha(x)$	case 1
3.	$\alpha(a)$	2, UI
4.	$\alpha(a) \vee \beta(a)$	3, I 1
5.	$(\forall x)\beta(x)$	case 2
6.	$\beta(a)$	5, UI
7.	$\beta(a) \vee \alpha(a)$	6, I 1
8.	$\alpha(a) \vee \beta(a)$	7, E 10
9.	$\alpha(a) \vee \beta(a)$	by cases
10.	$(\forall x)(\alpha(x) \vee \beta(x))$	9, Gen

Hence, $\vdash (\forall x)\alpha(x) \vee (\forall x)\beta(x) \Rightarrow (\forall x)(\alpha(x) \vee \beta(x))$

2.2 j

P1: $(\forall x)(\alpha(x) \Leftrightarrow (\exists y)\beta(y, x))$

P2 $(\forall x)(\forall y)(\beta(x,y) \Rightarrow (\gamma(x) \wedge \gamma(y)))$

Prove P1, P2 $\vdash (\forall x)(\alpha(x) \Rightarrow \gamma(x))$

1 $(\forall x)(\alpha(x) \Leftrightarrow (\exists y)\beta(y,x))$ from Γ

2 $(\forall x)(\forall y)(\beta(x,y) \Rightarrow (\gamma(x) \wedge \gamma(y)))$ from Γ

3 $\alpha(a)$ hypothesis

4 $\alpha(a) \Leftrightarrow (\exists y)\beta(y,a)$ 1, UI

5 $(\alpha(a) \Rightarrow (\exists y)\beta(y,a)) \wedge ((\exists y)\beta(y,a) \Rightarrow \alpha(a))$ 4, EI (8

6 $\alpha(a) \Rightarrow (\exists y)\beta(y,a)$ 5, I2

7 $(\exists y)\beta(y,a)$ 3, 6, I3

8 $\beta(c,a)$ 7, EI

9 $\beta(c,a) \Rightarrow (\gamma(c) \wedge \gamma(a))$ 2, UI (twice)

10 $\gamma(c) \wedge \gamma(a)$ 8, 9, I3

11 $\gamma(a)$ 10, I2

12 $\alpha(a) \Rightarrow \gamma(a)$

13 $(\forall x)(\alpha(x) \Rightarrow \gamma(x))$ 12, Gen

Hence P1, P2 $\vdash (\forall x)(\alpha(x) \Rightarrow \gamma(x))$

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2.3 (f)

P1: $(\forall x)(\forall y)((F(x) \wedge S(y)) \Rightarrow D(x,y))$

P2: $\sim(\exists x)(\exists y)(F(x) \wedge J(y) \wedge D(x,y))$

P3: $(\exists x)F(x)$

C: $(\forall y)(S(y)) \Rightarrow \sim J(y))$

let a be arbitrary constant, c new const.

1	$(\forall x)(\forall y)((F(x) \wedge S(y)) \Rightarrow D(x,y))$	from Γ
2	$\sim(\exists x)(\exists y)(F(x) \wedge J(y) \wedge D(x,y))$	from Γ
3	$\exists(x)\ F(x)$	from Γ
4	$F(c)$	3, EI
5	$S(a)$	hypothesis
6	$J(a)$	hypothesis
7	$F(c) \wedge S(a)$	4,5, I 6
8.	$(F(c) \wedge S(a)) \Rightarrow D(c,a)$	1, UI (twice)
9.	$D(c,a)$	7,8, I 3
10	$F(c) \wedge J(a)$	4,6, I 6
11	$(F(c) \wedge J(a)) \wedge D(c,a)$	10,9, I 6
12	$(\exists y)((Fc) \wedge J(y)) \wedge D(c,y))$	11, EQ

13 $(\exists x)(\exists y)((F(x) \wedge J(y)) \wedge D(x,y))$ 12, EQ

14 contradiction 2, 13

15 $\sim J(a)$

16 $S(a) \Rightarrow \sim J(a)$

17. $(\forall y)(S(y) \Rightarrow \sim J(y))$ 16, Gen

Hence, $P1, P2, P3 \vdash (\forall y)(S(y) \Rightarrow \sim J(y))$

2.3 (j)

$P1: (\forall x)(R(x,x) \Rightarrow (\forall y) R(x,y))$

$P2: (\forall x)(\forall y)(R(x,y) \Rightarrow R(y,x))$

$P3: (\forall x)(\exists y) \sim R(x,x)$

$C: (\forall x) \sim R(x,x)$

let a be an arbitrary constant, b new constant

1 $(\forall x)(R(x,x) \Rightarrow (\forall y) R(x,y))$ From Γ

2 $(\forall x)(\forall y)(R(x,y) \Rightarrow R(y,x))$ From Γ

3 $(\forall x)(\exists y) \sim R(x,x)$ From Γ

4 $(\exists y) \sim R(y,a)$ 3, UI

5 $\sim R(b,a)$ 4, EI

6	$R(a,a)$	hypothesis
7	$R(a,a) \Rightarrow (\forall y)R(a,y)$	1, UI
8	$(\forall y)\, R(a,y)$	6, 7, I3
9	$R(a,b)$	8, UI
10	$R(a,b) \Rightarrow R(b,a)$	2, UI (twice)
11	$R(b,a)$	9, 10, I3
12	contradiction	5, 11
13	$\sim R(a,a)$	proof by contr.
14	$(\forall x) \sim R(x,x)$	13 Gen

Hence $P1, P2, P3 \vdash C$

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2.4(g)

let predicate be defined:

$W(x)$: x is a writer

$U(x)$: x understand humon natur

$C(x)$: x is clever

$T(x)$: x is true poet

$H(x)$: x can stir hearts of men

$M(x)$: x wrote Hamlet

s : Shakespeare

$P_1 : (\forall x)((W(x) \wedge U(x)) \Rightarrow C(x))$

$P_2 : (\forall x)(T(x) \Rightarrow H(x))$

$P_3 : W(s) \wedge M(s)$

$P_4 : (\forall x)((W(x) \wedge \sim U(x)) \Rightarrow \sim H(x))$

$P_5 : (\forall x)(M(x) \Rightarrow T(x))$

$C : C(s)$

1	$(\forall x)((W(x) \wedge U(x)) \Rightarrow C(x))$	from Γ
2	$(\forall x)(T(x) \Rightarrow H(x))$	from Γ
3	$W(s) \wedge M(s)$	from Γ
4	$(\forall x)((W(x) \wedge \sim U(x)) \Rightarrow \sim H(x))$	from Γ
5	$(\forall x)(M(x) \Rightarrow T(x))$	from Γ
6	$W(s)$	3, I2
7	$M(s)$	3, I2
8	$M(s) \Rightarrow T(s)$	5, UI
9	$T(s)$	7, 8, I3

10	$T(s) \Rightarrow H(s)$	2, UI
11	$H(s)$	9, 10, I3
12	$(W(s) \wedge \sim U(s)) \Rightarrow \sim H(s)$	4, UI
13	$\sim U(s)$	hypothesis
14	$W(s) \wedge \sim U(s)$	6, 13, I6
15	$\sim H(s)$	12, 14, I3
16	$H(s) \wedge \sim H(s)$	11, 15, I6
17	$U(s)$	13-16, contradiction
18	$W(s) \wedge U(s)$	6, 17, I6
19	$(W(s) \wedge U(s)) \Rightarrow C(s)$	1, UI
20	$C(s)$	18, 19, I3

Hence $P1, P2, P3, P4, P5 \vdash C(s)$